

A reservoir rock wettability modifier for ultra-high temperature and high salinity environment

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ABSTRACT

During drilling process, the water phase in drilling fluids infiltrates rock fractures through capillary action. The surface wettability of dolomite is governed by multiple factors, resulting in an unstable wetting state. Studies have shown that altering the surface wettability of reservoir rocks to an intermediate wetting state can effectively reduce the damage of drilling fluids to oil and gas reservoirs and improve oil and gas recovery. Therefore, it is necessary to develop a reservoir protectant to prevent the water phase in the drilling fluid from intruding into the oil and gas reservoirs. Given this, a modified polysiloxane was synthesized to alter the surface wettability of dolomite. Tetramethylcyclotetrasiloxane (D^H₄) and octamethylcyclotetrasiloxane (D₄) were ring-opened copolymerized to obtain the hydrogen-containing polysiloxane, which in turn reacted with unsaturated hydrocarbons to obtain the modified polysiloxane. The ability of reservoir protectants to regulate the surface wettability of dolomite under high-temperature and high-salinity conditions was tested. The experimental results show that the reservoir protectant is able to alter the wettability of the dolomite surface to an intermediate wetting state by adsorption on the rock surface even after 16 h of aging at 240 °C and 15% salt concentration.

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1. Introduction

At present, more than half of the proven oil and gas are stored in carbonate reservoirs, with dolomite reservoirs accounting for over 50% of the carbonate rock reserves (Ghosh et al., 2020; Li et al., 2019). Therefore, with the increasing demand for oil and gas resources at present, the exploitation of oil and gas resources in dolomite reservoirs, especially deep dolomite reservoirs, is becoming more and more important.

During drilling operations, the high-porosity, high-permeability dolomite promotes the intrusion of liquid phases at rock surface (Karimi et al., 2012), leading to changes in wettability across different zones. These zones show either oil wettability, water wettability, or mixed oil-water wettability (Zhao et al., 2023). The term “wettability” is defined as “the tendency of a fluid to spread over or adhere to a solid surface in the presence of other immiscible fluids” (Golabi et al., 2012a,b,c). Wettability is one of the key parameters governing the petrophysical and

petrochemical properties of reservoir rocks (Hosseini et al., 2024), significantly impacting the development efficiency of oil and gas reservoirs (Zhao et al., 2025). Reservoir wettability is regulated by multiple factors (Golabi et al., 2012a,b,c; Hosseini et al., 2015), and common wettability modifiers include surfactants (Golabi et al., 2024; Noruzi et al., 2024), ionic liquids (ILs) (Dahham et al., 2023a; Khosravani et al., 2024; Moradi et al., 2024), nanoparticles (NPs) (Noruzi et al., 2024), and biosurfactants (Golabi et al., 2012a,b,c). Research indicates that saturated carbonate solutions can also alter reservoir wettability (Zaker et al., 2024).

During drilling operations, the water phase in drilling fluids readily infiltrates dolomite reservoirs via capillary action, thereby damaging hydrocarbon reservoirs and impairing enhanced oil recovery (EOR) (Maskari et al., 2020; Rellegadla et al., 2021; Yuan et al., 2014). Therefore, it is necessary to develop a reservoir protective agent to prevent the water phase in the drilling and completion fluids from invading oil and gas reservoirs. This is an important research idea to protect oil and gas reservoirs by altering the surface wettability of dolomite (Li et al., 2018; Seyyedi and Sohrabi, 2016). Altering the surface wettability of dolomite to intermediate wetting state can effectively reduce the capillary force and thus reduce the invasion of water phase, and the

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addition of amphiphobic material can also be conducive to maintaining wellbore stability and protecting oil and gas reservoirs (Das et al., 2018; Kawelah et al., 2020; Ravari et al., 2011; Sakthivel, 2021). At present, many researchers have been committed to enhancing oil and gas recovery rates by altering the wettability of rock surfaces.

The team led by Anita Seethepalli's (Seethepalli et al., 2005) investigated the change in wettability of calcite surfaces by the interaction of dilute alkaline anionic surfactant solutions with crude oil on carbonate mineral surfaces. The experimental results show that this method can alter the wettability of calcite surface to an intermediate wetting state at 25 °C. In the process of enhancing oil recovery, traditional surfactants cannot withstand the harsh salinity and temperature conditions (Abbood et al., 2023; Mehdipour and Hosseini, 2024). However, ILs exhibit high tolerance to harsh salinity and temperature conditions (Abbood et al., 2024).

The team led by Seyednooroldin Hosseini discovered through experiment that ILs can approach rock surfaces more effectively and accumulate readily on them. By doing so, they disrupt the bonds between asphaltene molecules and the rock, displace these molecules, and alter the wettability toward more water-wet state (Dahham et al., 2023b). Additionally, the team measured the interfacial tension between ILs and various crude oils at room temperature and across a range of salinities (0.15%–0.5%) and pH (3.5–11) conditions, demonstrating their ability to modulate the rock surface wettability (Moradi et al., 2024). Colloidal silica nanofluids were synthesized using sodium silicate for altering dolomite wettability from oil-wet to water-wet, thereby enhancing oil recovery (Zadeh et al., 2024). The team led by M.F. Pinto (Pinto et al., 2020) reported that water-soluble PEG-H (glycerophosphate) could be used as a carbonate wettability improver to improve the hydrophilicity of carbonate rocks. The surface wettability of carbonate rocks is altered to an intermediate wetting state, which effectively improves oil recovery. The team led by Ibraheem Salaudeen (Salaudeen et al., 2021) reported an experiment with silica nanoparticles dispersed in brine and nanofluids prepared by high-speed stirring and ultrasonic treatment to alter the wettability of limestone surfaces, and the results show that the prepared nanofluids have the effect of changing the wettability of limestone surface and altering the wettability of limestone surface to the intermediate wetting state when the salt concentration is 0.5%. The team led by Mohammed Sayed (Sayed and Al-Muntasheri, 2015) developed a treatment formulation based on fluorinated polymer surfactants, by which the wettability of the sandstone surface can be altered from strong water wettability to intermediate wettability at 150 °C, improving the relative permeability of condensate oil and natural gas, and effectively reducing the influence of condensate accumulation. All the above solutions can effectively alter the wettability of the reservoir rock surface, as a way to protect the reservoir and improve the oil and gas recovery. Although remarkable achievements have been made in the above studies, none of them considered the high temperature environment above 200 °C. Since a considerable portion of the dolomite reservoirs are located in deep strata in a harsh environment of high temperature and high salinity, it is urgent to formulate a reservoir protection agent that can be used in high-temperature and high-salt environment for drilling and completion. However, this topic has been rarely discussed.

Polysiloxanes are a class of polymers with silicon atoms and oxygen atoms as the skeleton (Majumdar et al., 2011; Shen et al., 2020). Due to the large energy of Si-O bond (460.5 kJ/mol), polysiloxanes have high thermal and chemical stability. In addition, various groups can be introduced into the main chain of

polysiloxanes through the addition reaction of Si-H bonds and unsaturated hydrocarbon, giving polysiloxanes different physico-chemical properties (Drake et al., 2011; Li et al., 2017). In this paper, a modified polysiloxane was developed as a regulator for dolomite surface wettability, which can regulate the surface wettability of dolomite to an intermediate wettability state at a temperature of 240 °C and a salt concentration of 15%, and the regulation mechanism of this wettability regulator on the surface wettability of dolomite was investigated herein.

2. Experiments

2.1. Materials

The commercial products used in this study were purchased from Titan Corporation, including D₄ (RG, Shanghai Titan), D^H₄ (RG, Shanghai Titan), hexamethyldisiloxane (RG, Shanghai Titan), Purolite® CT175 (AR, Shanghai McLean), chloroform (AR, Chengdu Cologne), dimethyl sulfoxide (AR, Chengdu Cologne), sodium allyl sulfonate (AR, Chengdu Cologne), platinum (0)-1,3-Divinyl-1,1,3,3-tetramethyldisiloxane (RG, Shanghai Titan), and dolomite (Sichuan Basin).

2.2. Synthesis of hydrogen-containing polysiloxanes

Purolite® CT175 (0.122 g, 1 wt%), D^H₄ (3.392 g, 0.014 mol), and hexamethyldisiloxane (0.8098 g, 0.005 mol) were added to an oven-dried 50 mL Schlenk tube under argon. The reaction was heated to 65 °C, and then D₄ (8.899 g, 0.03 mol) was added dropwise under argon. The reaction continued to stir at 65 °C for 12 h. After the reaction was completed, insoluble impurities were removed by filtration to obtain clarified oil.

The synthetic route is shown in Fig. 1. The Si-O bond between D₄ and D^H₄ occurs under the initiation of hydrogen ions to form active terminal groups, thus leading to chain growth. After growing to a certain chain length, the chain growth center is consumed by the capping agent, thus ending the chain growth process.

2.3. Optimization of aggregation conditions

The temperature resistance of a polymer exhibits a positive correlation with its molecular mass, which therefore serves as a critical factor in defining its thermal stability. Therefore, increasing the molecular mass of the polymer is an effective strategy to enhance its thermal stability.

The optimized reaction conditions were determined by single-factor experiments. The molecular mass of hydrogen-containing polysiloxanes like PDMS-co-PHMS was used as the evaluation standard, and the relative viscosity (t_1/t_0 , where t_1 is the flow time of the polymer solution in the Ubbelohde viscometer, and t_0 is the flow time of pure solvent in the Ubbelohde viscometer) of the polymer solution was applied to characterize the molecular mass of hydrogen-containing polysiloxanes, including the PDMS-co-PHMS. The aggregation conditions were investigated, including reaction temperature, reaction time, the amount of end-capping agent (MM) and initiator (Purolite® CT175).

Initially, the polymerization was carried out at 65 °C for 12 h. In this system, the initiator of 1 wt% was employed, and the molar ratio of monomers (D₄: D^H₄: MM) was fixed at 6 : 2 : 1.

2.4. Synthesis of modified polysiloxanes

A mixture of sodium allyl sulfonate (95.1 mg) and dimethyl

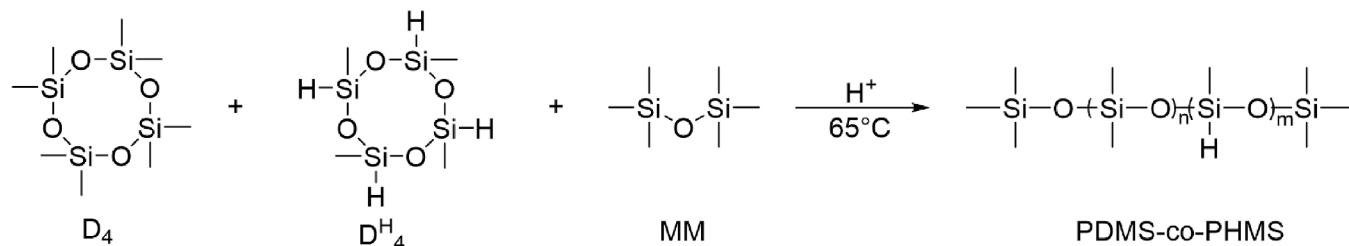


Fig. 1. Synthetic route of hydrogen-containing polysiloxanes.

sulfoxide (0.5 mL) was placed in an oven-dried 50 mL Schlenk tube, which was then purged with an argon atmosphere. Subsequently, chloroform (5 mL) and platinum(0)-1,3-divinyl-1,1,3,3-tetramethyldisiloxane (50 μL) were added to the reaction mixture. The mixture was heated to 85 $^\circ\text{C}$ and maintained at this temperature for 15 min. Subsequently, in-situ hydrogen-containing polysiloxanes (1 g) were added dropwise under an argon atmosphere. The Schlenk tube was then sealed, and the reaction was stirred at 80 $^\circ\text{C}$ for 36 h. Upon completion, the crude product was purified by filtration to remove insoluble impurities, followed by rotary evaporation to obtain white oily liquid. The synthetic route is depicted in Fig. 2, and the structure of the obtained product was confirmed by Proton Nuclear Magnetic Resonance ($^1\text{H-NMR}$) spectroscopy and elemental analysis (EA).

2.5. Performance testing

2.5.1. Aging treatment

The sulfonic acid group-modified polysiloxanes and deionized water were mixed at a mass ratio of 1 : 70 in a hydrothermal kettle. The mixture was then subjected to an aging treatment at 240 $^\circ\text{C}$ under a salt concentration of 15% for 16 h. After the aging process ended, the resulting solution was collected for subsequent use.

2.5.2. Wettability test

The wettability of the dolomite surface was evaluated by contact angle measurements. Dolomite flakes, with both ground and polished surfaces, were immersed in the aging solution. The contact angle of industrial white oil (No.5) on the lower surface of the treated dolomite was measured using the interface an integrated interface parameter measurement system (KRUS DSA30S).

2.5.3. Adsorption capacity determination

The absorbance of polymer solutions at various concentrations was measured using a UV-759 spectrophotometer. A calibration curve of absorbance versus concentration was established based on the Lambert-Beer law. The absorption amount of the reservoir protection agent on the rock surface was then calculated by measuring the absorbance of the solution before and after the adsorption process.

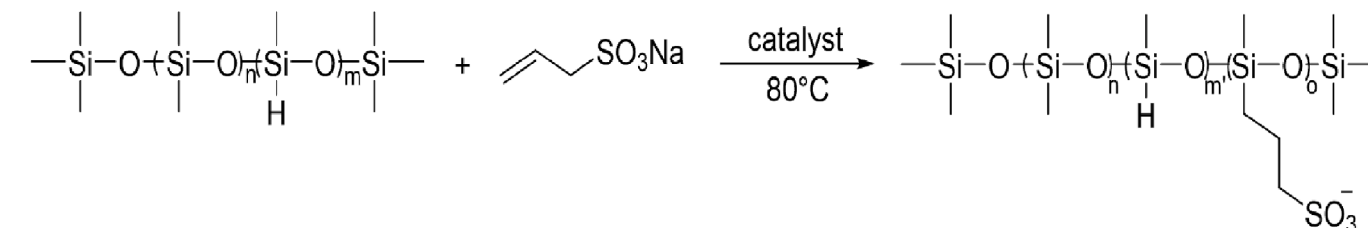


Fig. 2. Synthetic routes of sulfonic acid group-modified polysiloxanes.

3. Results and discussion

3.1. Proton Nuclear Magnetic Resonance

Figure 3 shows the Proton Nuclear Magnetic Resonance ($^1\text{H-NMR}$) spectrum of the hydrogen-containing polysiloxanes (e.g., PDMS-co-PHMS). The characteristic resonances appear at 4.71 ppm and in the range of 0.09–0.12 ppm, which are assigned to the Si-H and Si-CH₃ protons, respectively. The Si-CH₃ signal region reveals methyl groups at distinct locations: the chain-terminal methyl groups (a) and the main-chain middle methyl groups (b). These chemical shifts agrees well with literature data (Cheng et al., 2014).

The $^1\text{H-NMR}$ spectrum in Fig. 4 confirms the successful modification of polysiloxanes with sulfonic acid groups. The following signals were identified: the Si-H proton at 4.70 ppm, the Si-CH₂ (c) protons at 0.89 ppm, the middle methylene units Si-CH₂-CH₂ (d) at 1.26 ppm, and the terminal methylene protons H (e) in the propyl chain (Si-CH₂-CH₂-CH₂) at 1.38 ppm.

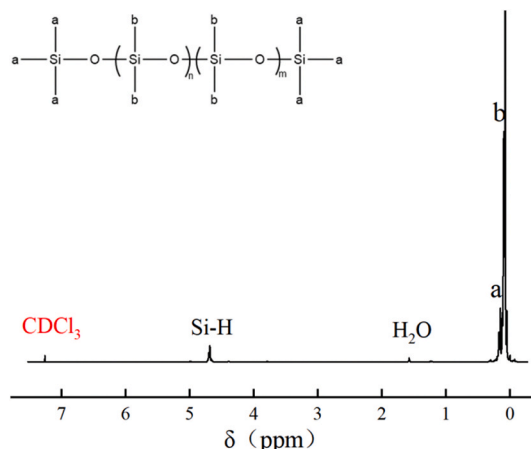


Fig. 3. $^1\text{H-NMR}$ spectrum of hydrogen-containing polysiloxanes like PDMS-co-PHMS. The "a" is the chain-terminal methyl groups; "b" denotes the main-chain middle methyl groups; and δ represents chemical shifts.

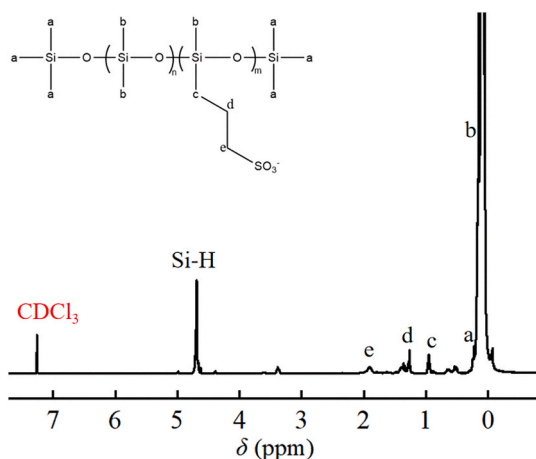


Fig. 4. ^1H -NMR spectrum of sulfonic acid group-modified polysiloxanes. The “a” is the chain-terminal methyl groups; “b” is the main-chain middle methyl groups; “c” is the Si-CH protons; “d” is the middle methylene units Si-CH₂-CH₂; “e” is the terminal methylene protons H; and δ represents chemical shifts.

3.2. Results of molecular mass optimization

3.2.1. The influence of reaction temperature on polymer molecular mass

The optimum reaction temperature was fixed under the initial conditions set at 0.05 mmol (0.5 Equiv) of hexamethyldisiloxane (3a), 1 wt% of initiator and a reaction time of 12 h. As shown in Fig. 5a, the reaction temperature has a great influence on the polymer molecular mass. An increase in temperature generally promoted monomer conversion, as evidenced by the rise in molecular mass. The dichloromethane solution of the polymer synthesized at 85 °C exhibited the greatest relative viscosity, indicating the maximum molecular mass achieved under these conditions. However, a further increase in temperature led to an excessive rise in the viscosity of the reaction system, which impeded efficient monomer polymerization. So, 85 °C was identified as the optimal reaction temperature.

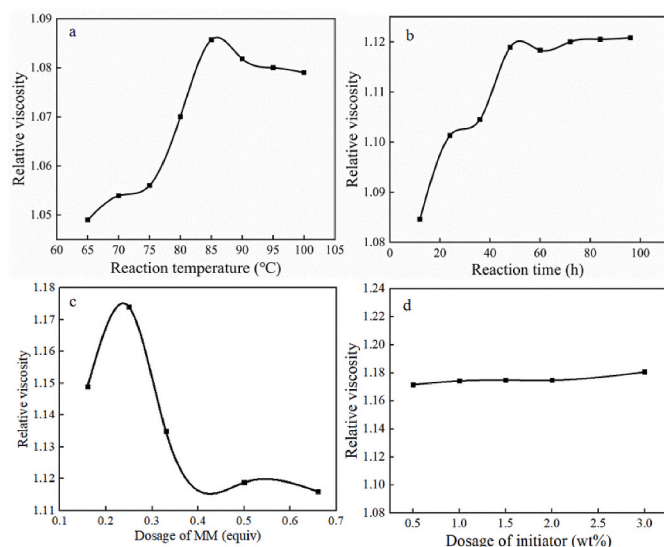


Fig. 5. Screening of reaction temperature (a), reaction time (b), capping-agent (MM) dosage (c), and initiator dosage (d).

3.2.2. The influence of reaction time on polymer molecular mass

As shown in Fig. 5b, the reaction temperature was altered to 85 °C based on the initial conditions to identify the reaction time. An increase in the reaction time is also beneficial to the polymer molecular mass rise. When the reaction time was extended from 12 h to 48 h, the relative viscosity exhibited an excessive rise. However, with the reaction time continuing to increase, the relative viscosity increased slightly. Therefore, the reaction time should not be too long for practical reasons, and 48 h was identified as the optimum.

3.2.3. The influence of the amount of capping agent MM on polymer molecular mass

Building on the optimized reaction conditions of 85 °C and 48 h, the amount of chain-terminating agent or capping agent (MM) was screened. As shown in Fig. 5c, a series of MM loadings—0.16, 0.25, 0.33, 0.5 and 0.66 equivalents relative to 1 equivalent of D₄—were evaluated. An excess of MM prematurely terminates the growing polymer chains, resulting in a lower molecular mass. Conversely, an insufficient amount of MM leads to a loss of molecular mass control and a sharp increase in system viscosity, which also hinders chain propagation. Based on the experimental results, 0.25 equivalents of MM was identified as the optimal loading, yielding the polymer with the highest molecular weight.

3.2.4. The influence of the Purolite® CT175 amount on polymer molecular mass

The initiator Purolite® CT175 loading was optimized under the initial conditions featuring a reaction temperature of 85 °C, a reaction time of 48 h, and a capping agent MM fixed at 0.25 equivalent (Fig. 5d). Using 1 equivalent of D₄ as the benchmark for the screening, the effect of initiator dosage on the polymer molecular mass was evaluated. The initiator amount was found to have a much smaller effect on molecular mass than the prior factors. Given the minimal impact observed, we selected 1 wt% Purolite® CT175 as the optimal dosage.

Regarding the screened conditions shown above, we determined the optimal synthesis conditions for hydrogen-containing polysiloxanes. The Gel Permeation Chromatography (GPC) results of the polymer, synthesized before and after process optimization, are summarized in Table 1. The data show a significant increase in the number-average molecular weight (M_n) of from 3003 to 7782, demonstrating the effectiveness of the optimized conditions.

3.3. Elemental analysis

The elemental analysis (EA) results are shown in Table 2. We took two samples of different quality for testing, and the results showed that there was about 3.5% sulfur in the molecule for both of them. Combined with the ^1H -NMR spectrum analysis, these results collectively demonstrated the successful introduction of sulfonic acid groups into the polysiloxanes molecular chains as designed.

Table 1

Molecular mass statistics of hydrogen-containing polysiloxanes before and after reaction condition optimization.

Distribution type	M_n	M_w	M_z	M_w/M_n
Before optimization	3003	5361	10664	1.78
After optimization	7782	15439	30720	1.98

Table 2
Elemental analysis data of sulfonic acid group-modified polysiloxanes.

Weight (mg)	C (%)	N (%)	H (%)	S (%)
7.5780	29.42	0.11	6.702	3.532
10.9270	29.39	0.10	6.763	3.487

Table 3
Porosity of the artificial dolomite cores.

Core labelling	Core specification	Porosity (%)
10–19	Dia. 2.5 cm * L 5 cm	10.34
10–21	Dia. 2.5 cm* L 5 cm	9.86

Notes: "Dia." represents diameter, and L is length.

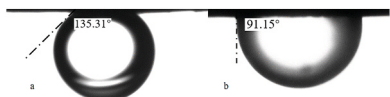


Fig. 6. Schematic diagram of the three-phase contact angle between dolomite surface and different solution. (a) Contact angle on dolomite surface in saline solution with a concentration of 15%; and (b) contact angle on dolomite surface in a sulfonic acid-modified polysiloxane aging solution.

Table 4
The effect of polymer concentration on the contact angle adjustment ability of rock surface.

Concentration (%)	0	0.2	0.3	0.4	0.5
Contact angle (°)	135.31	116.89	92.14	89.56	91.15

Table 5
The effect of salt concentration on wettability adjustment ability.

Concentration (%)	0	5	10	15
Contact angle (°)	97.20	96.74	94.57	89.56

Table 6
The effect of aging temperature on wettability adjustment ability.

Temperature (°C)	180	220	240	250
Contact angle (°)	88.12	86.94	89.56	90.15

Table 7
Adsorption data of different concentrations of aging solution on rock surface.

Solution quantity (%)	0.05	0.10	0.15	0.20
Adsorption capacity (10 ⁻² mg/cm ²)	3.68	7.58	10.88	12.87

3.4. Dolomite core porosity

The porosity of two artificial dolomite cores was evaluated with Model DP-DYX-6G, a core porosity tester. The results are presented in Table 3.

3.5. Ability to optimize wettability

Figure 6a depicts the contact angle of industrial white oil (No.5) on a dolomite surface tested in saline solution with a concentration of 15%. The measured contact angle is 135.31°, showing a hydrophilic state. Figure 6b shows the contact angle of industrial white oil (No.5) on a dolomite surface treated with the aging

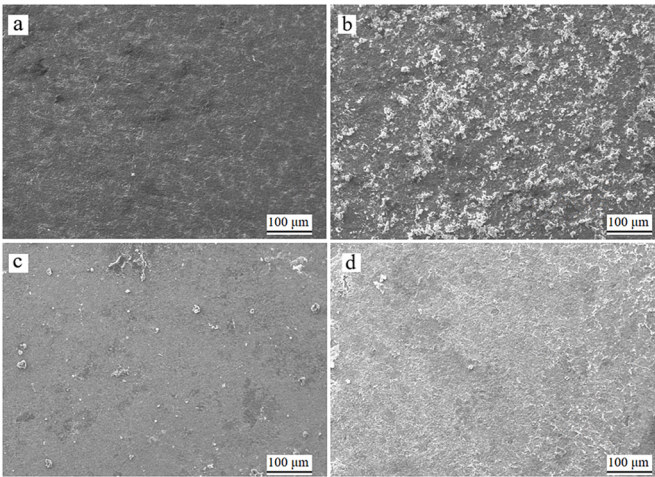


Fig. 7. SEM images of dolomite samples immersed in aging solutions of varying polymer concentrations. (a) 0; (b) 0.2%; (c) 0.3%; and (d) 0.4%.

solution of sulfonic acid-modified polysiloxanes. The measured contact angle is 91.15°, exhibiting an intermediate wettability state. Experimental results indicate that the modified polysiloxane aging solution can indeed alter the wettability of the dolomite surface to an intermediate wetting state when the contact angle ranges from 75° to 105°.

3.5.1. Factors influencing the performance of the wettability modifier

To systematically investigate the factors affecting the performance of the wettability modifier, this study evaluated its ability to regulate the surface wettability of dolomite under different dosages, salinities, and aging temperatures. The results are summarized in Tables 4–6. To examine the influence of polymer concentration on wettability modification, five polymer solutions with gradient concentrations were aged in a 15% brine environment at 240 °C for 16 h. After aging, the wettability of dolomite in each solution was measured. Table 4 shows that as the polymer concentration decreases, the wettability of the dolomite surface shifts toward hydrophilic behavior. The ability of the aged polymer solution to modify dolomite wettability is clearly concentration-dependent within the range of 0–0.4%. When the polymer concentration is reduced to 0.2%, a significant decline in wettability alteration capability is observed. It is hypothesized that below a certain threshold concentration, the polymer cannot form a complete adsorption film on the rock surface, hereby losing its ability to effectively regulate rock wettability. At a polymer concentration of 0.5%, the measured contact angle shows little difference from the result at 0.4%. In practical applications, reducing the dosage of additives can enhance economic benefits (see Table 7).

To verify the above conclusions, five sets of clean dolomite samples were immersed in aging solutions with concentrations of 0, 0.2%, 0.3%, 0.4% and 0.5% for 5 min. After removal and drying, the surface morphology of the samples was characterized using scanning electron microscopy. The results, as shown in Fig. 7, demonstrate that the amount of polymer adsorbed on the dolomite surface increases with the concentration of the aging solution. When the concentration reaches 0.3%, a continuous polymer film forms on the rock surface. This observation provides microscopic evidence that at a polymer concentration of 0.3%, a complete modified layer can be formed on the sample surface, thereby altering the wettability of the dolomite to an intermediate wetting

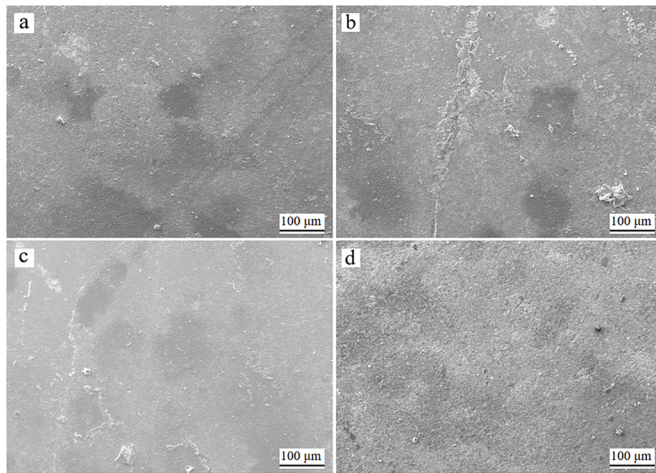


Fig. 8. SEM images of dolomite samples immersed in aging solutions of varying salt concentrations. (a) 0%; (b) 5%; (c) 10%; and (d) 15%.

state.

Table 5 summarizes the influence of salinity on wettability alteration performance. Under constant conditions of polymer concentration (0.4%) and aging temperature (240 °C), experiments were conducted at four different salinity levels. The data reveal that as the salinity increases, the hydrophilicity of the rock surface exhibits a gradually decreasing trend.

To investigate the underlying mechanism of the above experimental results, four clean dolomite samples were immersed in aging solutions with four different salinity levels for 5 min. After removal and drying, the surface morphology of the samples was characterized by scanning electron microscopy. The results are shown in Fig. 8. Given the minor differences in wettability alteration and the similar surface morphological features observed among the four samples, it can be concluded that the variation in rock surface wettability is not primarily governed by the polymer, but is mainly influenced by changes in Na⁺ concentration (Bind and Sharma, 2022).

Table 6 shows the effect of temperature on the wettability regulation performance of the wettability modifier. Experiments were conducted using a polymer solution with a concentration of 0.4% and a salinity of 15%, aged at four different temperatures (180 °C, 220 °C, 240 °C, and 250 °C) for 16 h. The results

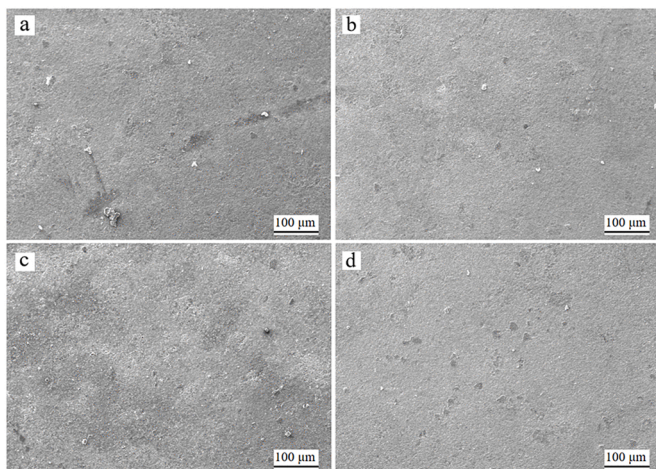


Fig. 9. Surface morphology of dolomite samples after 5 min immersion at four different aging temperatures. (a) 180 °C; (b) 220 °C; (c) 240 °C, and (d) 250 °C.

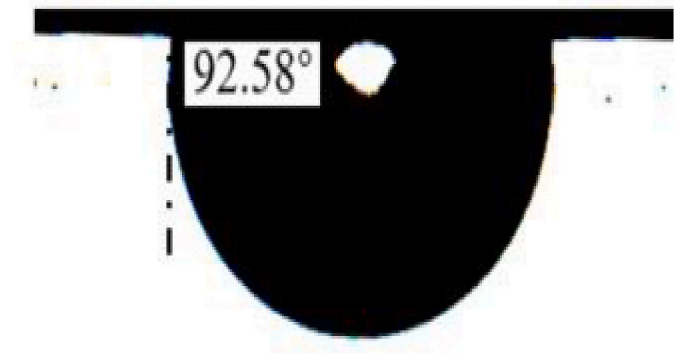


Fig. 10. A contact angle image of the polymer-treated dolomite surface.

demonstrate that variations in aging temperature have no significant impact on the ability of the polymer to alter the wettability of the rock surface. This indicates that within the experimental temperature range, the polymer maintains its structural stability, thus sustaining effective control over the rock surface wettability.

Figure 9 shows the surface morphology of rock samples after treatment at four different aging temperatures for 5 min. The characterization results show that there is no significant difference in the surface morphology of each rock sample, which is covered by a complete and continuous polymer film. Since other potential interference factors have been excluded from the experimental system, each rock sample exhibits similar surface wetting characteristics under different temperature conditions, indicating that the polymer has stable temperature resistance in the experimental temperature range (including 250 °C), and its wettability-controlling ability is not significantly affected by temperature changes.

3.5.2. Performance testing in simulated in-situ environments

To evaluate the practical performance of the wettability modifier under simulated deep reservoir conditions, experiments were designed to replicate downhole environments. Polished dolomite slices were placed in at HTHP filter press, and the aged wettability modifier solution was introduced. The solution was released at 240 °C to simulate the scouring action of drilling fluid on rock surfaces in high-temperature formations. The wettability of the

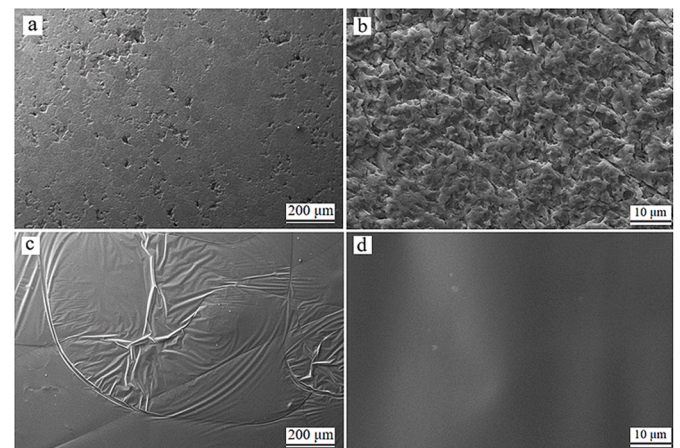


Fig. 11. SEM images showcasing the microstructures of dolomite samples before (a, b) and after (c, d) treatment. Images (b) and (d) reveal the corresponding areas in (a) and (c) at a 20 × higher magnification, respectively.

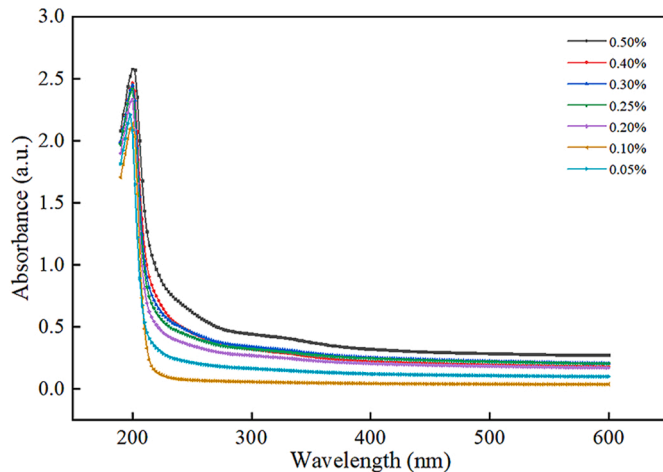


Fig. 12. Ultraviolet absorption curves of polymer aging solutions with different standard concentrations.

treated rock slices was subsequently measured in 15% brine, with the test results presented in Fig. 10.

The measured three-phase contact angle between the treated dolomite and No.5 industrial white oil in 15% brine was determined to be 92.58° , maintaining an intermediate wetting state. This suggests that the wettability alteration of dolomite is achieved through adsorption of the polymer constituents from the wettability modifier onto the rock surface.

To further investigate the underlying mechanism, comparative characterization of the surface morphology was conducted on dolomite samples before and after treatment. The microscopic structural variations are illustrated in Fig. 11.

Characterization results indicate that untreated rock samples exhibit a rough surface structure, and this roughness enhances the wettability of the wetting system (Pandey and Roy, 2013) (Fig. 11). As dolomite is intrinsically water wettability rock, increased

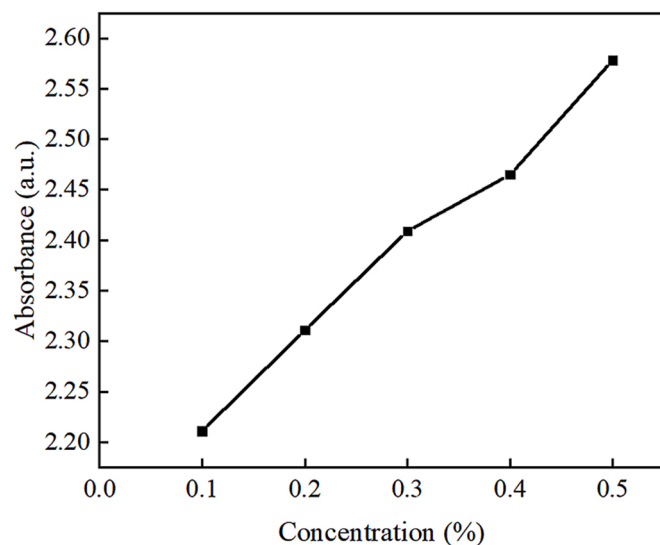


Fig. 13. Standard adsorption curve of ageing solution at polymer concentrations of 0.1–0.5.

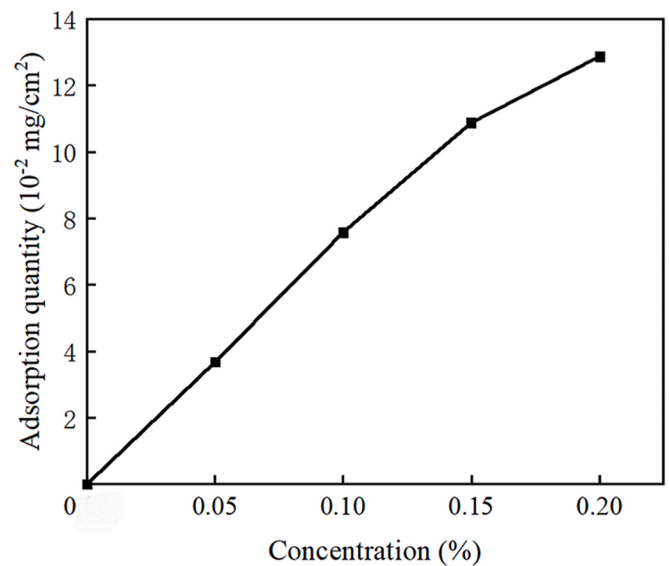


Fig. 14. Adsorption curves of different concentrations of aging solution on rock surfaces.

surface roughness further amplifies its hydrophilic characteristics. Comparative analysis of images (c) and (d) demonstrate that the treated rock surface is covered with an adsorbed polymer layer that substantially reduces surface roughness. It can therefore be inferred that the wettability modifier adsorbs onto the rock surface to form a polymer film, which significantly diminishes the surface roughness of dolomite, thereby reducing its hydrophilicity and effectively modulating the wettability.

3.6. Determination of polymer adsorption capacity on dolomite

The absorbances of a series of polymer solutions with varying concentrations were determined with a UV-Vis spectrophotometer, and a calibration curve was plotted (Figs 12 and 13).

The adsorption of polymer on the rock surface increases with the solution concentration, but the growth rate slows and the extent of increase narrows after the concentration reaches 0.2% (Fig. 14). This attenuation is likely attributable to the saturation of available surface sites or an alteration in the adsorption kinetics.

4. Conclusions

This study successfully synthesized sulfonic acid-modified polysiloxane, and its structure was characterized by $^1\text{H-NMR}$, gel permeation chromatography (GPC) and elemental analysis (EA). Experiments demonstrated that this polymer could effectively alter the wettability of the dolomite surface to intermediate wettability under harsh conditions of 240°C and 15% salinity, thereby contributing to reservoir protection. The effects of polymer concentration, salinity, and aging temperature on its performance were systematically investigated. UV-Vis spectrophotometric analysis of aged solutions at different concentrations revealed that polymer adsorption on the dolomite surfaces increased with concentration. Further SEM microanalysis indicated the mechanism: polymer molecules adsorb onto the rock surface to form a smooth film, significantly reducing surface roughness and consequently achieving wettability alteration.

CRediT authorship contribution statement

Ling Lin: Writing – original draft, Methodology, Investigation, Data curation. **Yulong Bai:** Writing – review & editing, Conceptualization. **Yue Feng:** Writing – review & editing, Supervision. **Yuanzhi Qu:** Funding acquisition. **Taigang Zhou:** Funding acquisition, Data curation.

Declaration of generative AI and AI-assisted technologies in the writing process

The generative AI and AI-assisted technologies were not employed by the authors during the authoring process.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yuanzhi Qu is currently employed by CNPC. This research project was funded by the Opening Project of Oil & Gas Field Applied Chemistry Key Laboratory of Sichuan Province (YQKF202214).

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